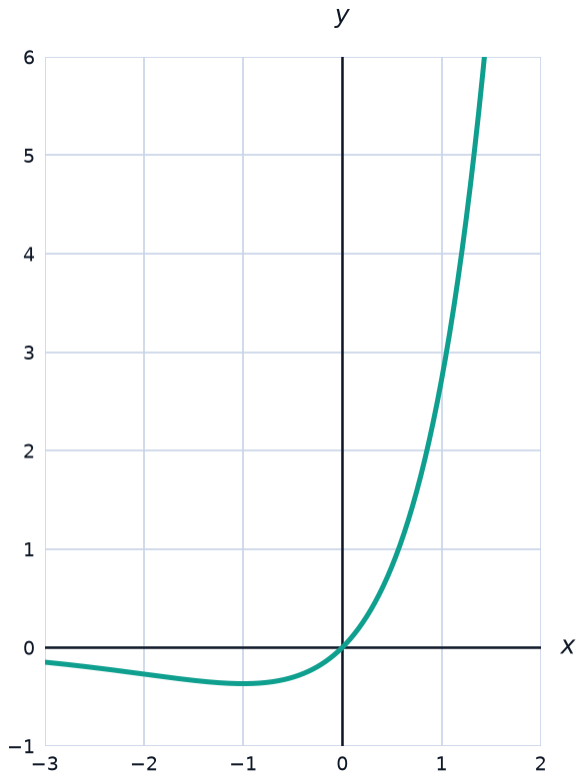
Integration by Parts

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

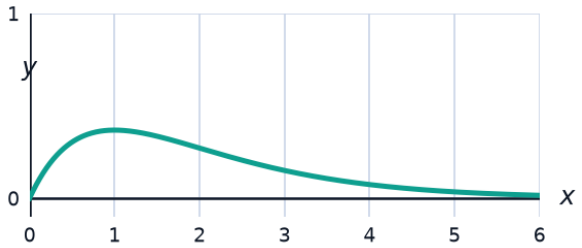
1 [Foundational] • Target: The integration by parts formula

Evaluate $\int xe^x\,dx$.



2 [Foundational] • Target: Definite integrals: area under a transcendental product

Find the exact area under $y = xe^{-x}$ on $[0, 3]$, using $\int xe^{-x}\,dx = -xe^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).



3 [Foundational] • Target: The integration by parts formula

State the integration by parts formula, and explain the LIATE guideline for choosing u.

4 [Foundational] • Target: The integration by parts formula

Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.

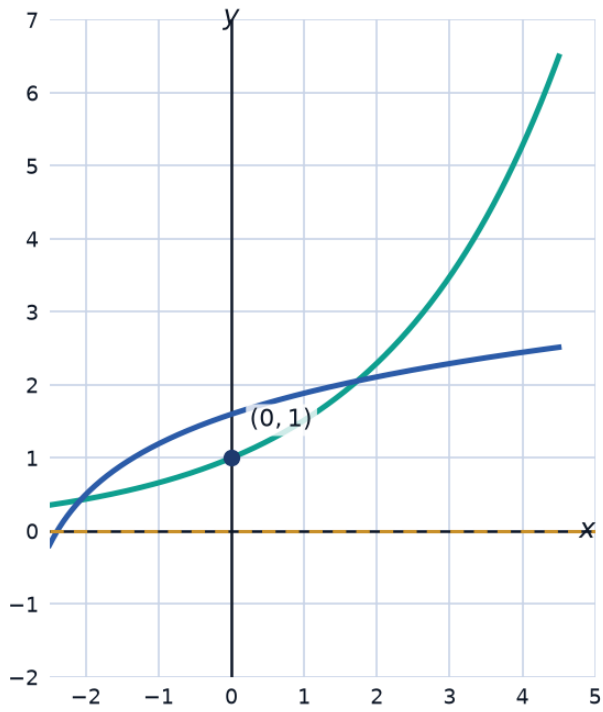
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5 [Foundational] • Target: The integration by parts formula

Evaluate $\int x\cos x\,dx$.

6 [Foundational] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int \ln x\,dx$.



7 [Foundational] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int x\ln x\,dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int \tan^{-1} x\,dx$.

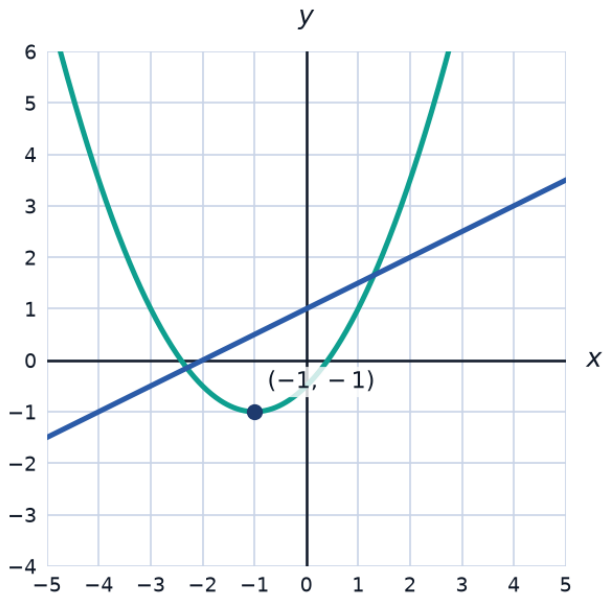
9 [Intermediate] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int x^2\ln x\,dx$.

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10 [Intermediate] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int \sin^{-1} x \, dx$.



11 [Intermediate] • Target: Repeated integration by parts

Evaluate $\int x^2 e^x \, dx$, applying integration by parts twice.

12 [Intermediate] • Target: Repeated integration by parts

Evaluate $\int x^2 \sin x \, dx$, applying integration by parts twice.

13 [Intermediate] • Target: Repeated integration by parts

Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x \, dx$, identifying how many times integration by parts must be applied.

14 [Intermediate] • Target: The cyclic case

Evaluate $\int e^x \sin x \, dx$ using the cyclic (solve-for-the-integral) technique.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

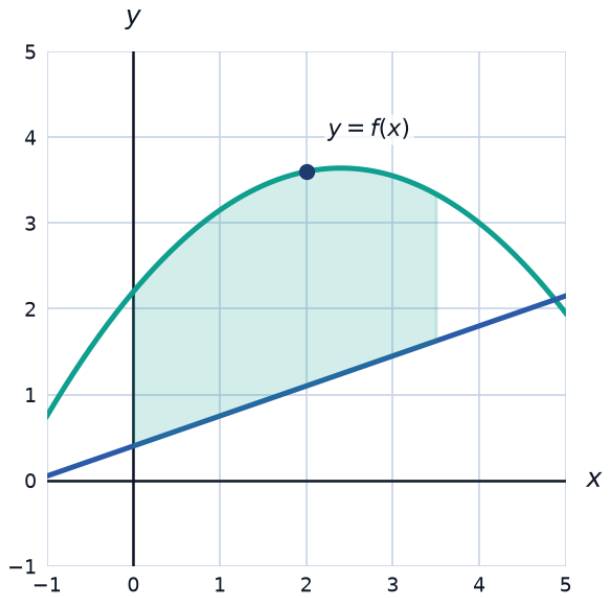
15 [Challenging] • Target: The cyclic case • Cross-Lesson: 4.01 Polar Curves and Derivatives

Evaluate $\int e^{2x} \cos x \, dx$ using the cyclic technique.

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16 [Challenging] • Target: Definite integrals using integration by parts • Cross-Lesson: 4.02 Area in Polar Coordinates

Evaluate $\int_0^1 xe^x\,dx$ using the antiderivative found earlier.



17 [Challenging] • Target: Definite integrals using integration by parts • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Evaluate $\int_1^e \ln x\,dx$ using the antiderivative found earlier.

18 [Challenging] • Target: Definite integrals using integration by parts • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Evaluate $\int_0^\pi x\sin x\,dx$.

19 [Challenging] • Target: Integration by parts to derive a reduction formula • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Use integration by parts to show $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x\,dx$.

20 [Challenging] • Target: Choosing u and dv wisely • Cross-Lesson: 6.02 Taylor and Maclaurin Series

For each integral, state the best choice of u (using LIATE) without evaluating:

a $\int x^3 \ln x\,dx$

b $\int xe^{5x}\,dx$

c $\int x\tan^{-1}x\,dx$

d $\int x^2\cos x\,dx$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The integration by parts formula

Evaluate $\int_0^{\pi} -2x\cos x\,dx$, using integration by parts.

a. 2

b. 6

c. 4

d. -4

22 [Foundational] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int_0^{\pi} -6x\cos x\,dx$, using integration by parts.

a. 6

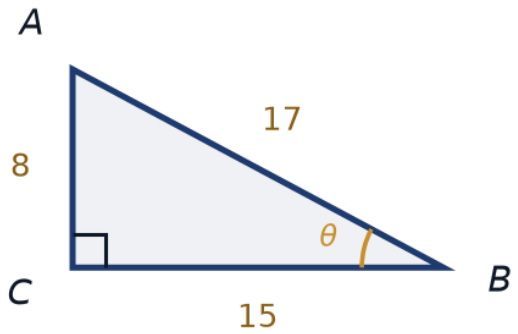
b. 14

c. 12

d. -12

23 [Foundational] • Target: Repeated integration by parts

Evaluate $\int_0^{\pi} -7x\cos x\,dx$, using integration by parts.



a. 16

b. 7

c. -14

d. 14

24 [Foundational] • Target: The cyclic case

Evaluate $\int_0^{\pi} 9x\cos x\,dx$, using integration by parts.

a. -16

b. 18

c. -18

d. -9

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25 [Foundational] • Target: Definite integrals using integration by parts

Evaluate $\int_0^{\pi} -x\cos x\,dx$, using integration by parts.

a. -2

b. 4

c. 2

d. 1

26 [Foundational] • Target: Integration by parts to derive a reduction formula

Evaluate $\int_0^{\pi} -3x\cos x\,dx$, using integration by parts.

a. 3

b. -6

c. 8

d. 6

27 [Foundational] • Target: Choosing u and dv wisely

Evaluate $\int_0^{\pi} -4x\cos x\,dx$, using integration by parts.

a. 4

b. -8

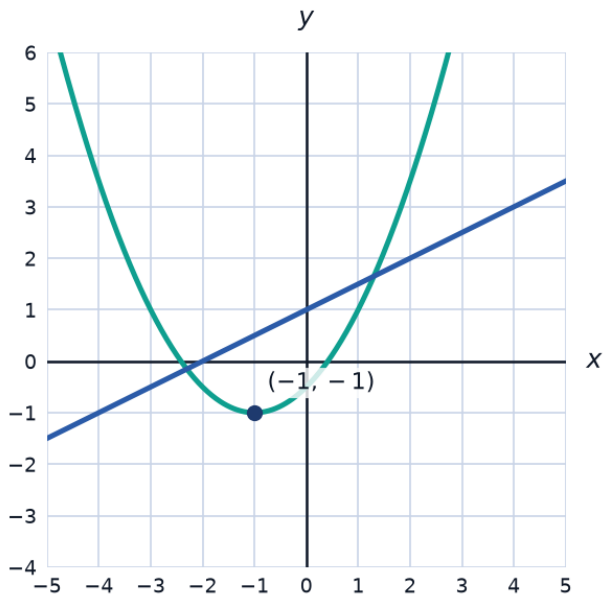
c. 10

d. 8

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28 [Foundational] • Target: Integration by parts with algebraic-trigonometric products

Given $f(0)=0$, $f(3)=2$, $g(0)=2$, $g(3)=2$, and $\int_0^3 f'(x)g(x)\,dx=-2$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.



a. 10

b. -6

c. 2

d. 6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Definite integrals: area under a transcendental product

Given $f(1)=5$, $f(3)=6$, $g(1)=4$, $g(3)=-6$, and $\int_1^3 f'(x)g(x)\,dx=-7$, use integration by parts to find $\int_1^3 f(x)g'(x)\,dx$.

a. -45

b. -49

c. -63

d. 49

30 [Intermediate] • Target: Mixed review: choosing the right technique

Given $f(3)=-1$, $f(6)=2$, $g(3)=1$, $g(6)=3$, and $\int_3^6 f'(x)g(x)\,dx=-8$, use integration by parts to find $\int_3^6 f(x)g'(x)\,dx$.

a. -1

b. 15

c. -15

d. 19

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31 **[Intermediate]** • **Target: Standard Integration by Parts Formula**

Given $f(3) = 5$, $f(7) = 0$, $g(3) = 6$, $g(7) = -5$, and $\int_3^7 f'(x)g(x)\,dx = -10$, use integration by parts to find $\int_3^7 f(x)g'(x)\,dx$.

a. -40

b. -20

c. 20

d. -16

32 **[Intermediate]** • **Target: Tabular Integration by Parts**

Given $f(0) = 5$, $f(3) = -2$, $g(0) = 3$, $g(3) = 1$, and $\int_0^3 f'(x)g(x)\,dx = -6$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.

a. -7

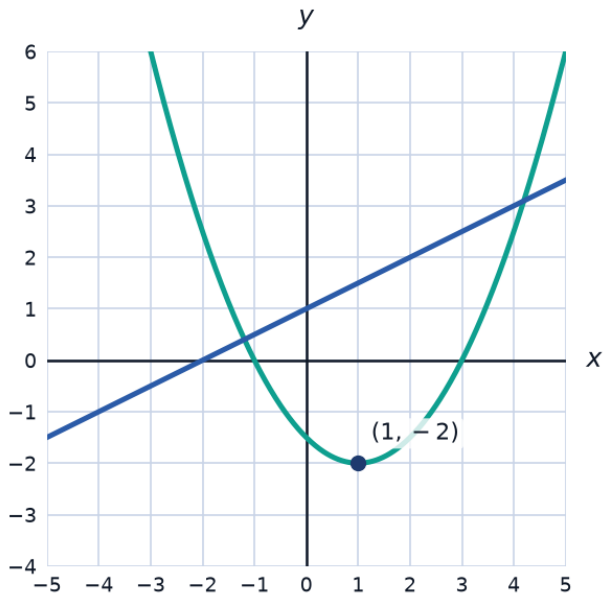
b. -11

c. -23

d. 11

33 **[Intermediate]** • **Target: Cycling Integrals (Solving for the Unknown Integral)**

Given $f(3) = -4$, $f(4) = 6$, $g(3) = -4$, $g(4) = -6$, and $\int_3^4 f'(x)g(x)\,dx = -2$, use integration by parts to find $\int_3^4 f(x)g'(x)\,dx$.



a. -46

b. -50

c. -54

d. 50

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34 **[Intermediate]** • **Target: Integration by Parts with Inverse Trigonometric Functions**

Given $f(0) = 0$, $f(3) = -6$, $g(0) = -3$, $g(3) = 4$, and $\int_0^3 f'(x)g(x)\,dx = -4$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.

a. -28

b. 20

c. -20

d. -16

35 **[Intermediate]** • **Target: The integration by parts formula**

Using the cyclic technique, $\int e^{3x}\cos(3x)\,dx = \frac{e^{3x}(3\cos(3x) + C_1\sin(3x))}{18} + C$. Find the coefficient C_1.

a. $\frac{1}{3}$

b. $-\frac{1}{6}$

c. $\frac{1}{6}$

d. $\frac{7}{6}$

36 **[Intermediate]** • **Target: Integration by parts with logarithms and inverse trig**

Using the cyclic technique, $\int e^{5x}\cos(4x)\,dx = \frac{e^{5x}(5\cos(4x) + C_1\sin(4x))}{41} + C$. Find the coefficient C_1.

a. $-\frac{4}{41}$

b. $\frac{4}{41}$

c. $\frac{5}{41}$

d. $\frac{4}{25}$

37 **[Intermediate]** • **Target: Repeated integration by parts**

For $\int x^2\tan^{-1}x\,dx$, which is the correct choice of u according to LIATE?

a. $u = \tan^{-1}x$ (Inverse Trig outranks Algebraic in LIATE)

b. $u = x^2$ (Algebraic always goes first)

c. It does not matter which factor is chosen for u

d. $u = x^2$ (the opposite LIATE priority)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The cyclic case • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For $\int x^2 e^{6x}\,dx$, which is the correct choice of u according to LIATE?

a. $u = x^2$ (Algebraic outranks Exponential in LIATE)

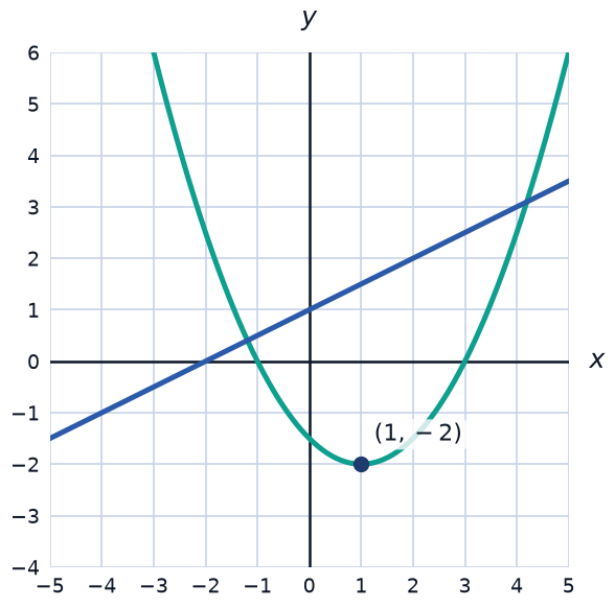
b. $u = x^2$ (Algebraic always goes first)

c. It does not matter which factor is chosen for u

d. $u = e^{6x}$ (the opposite LIATE priority)

39 [Challenging] • Target: Definite integrals using integration by parts • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Given $f(1) = 6$, $f(4) = 5$, $g(1) = 4$, $g(4) = 0$, and $\int_1^4 f(x)g'(x)\,dx = -14$, use integration by parts to find $\int_1^4 f'(x)g(x)\,dx$.



a. -6

b. -38

c. 10

d. -10

40 [Challenging] • Target: Integration by parts to derive a reduction formula • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Given $f(0) = 2$, $f(2) = -2$, $g(0) = 0$, $g(2) = -3$, and $\int_0^2 f(x)g'(x)\,dx = -10$, use integration by parts to find $\int_0^2 f'(x)g(x)\,dx$.

a. -4

b. 16

c. 20

d. -16

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41 [Challenging] • Target: Standard Integration by Parts Formula • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Evaluate $\int xe^{2x}dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$

b. $\frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + C$

c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{2}x^2e^{2x} + C$

42 [Challenging] • Target: Standard Integration by Parts Formula • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Evaluate $\int_0^1 x\cos(\pi x)dx$.

a. $-\frac{2}{\pi^2}$

b. $\frac{2}{\pi^2}$

c. 0

d. $-\frac{1}{\pi}$

43 [Challenging] • Target: Tabular Integration by Parts • Cross-Lesson: 3.01 Parametric Equations

When is tabular integration applicable?

a. When one factor is a polynomial that terminates to 0 upon repeated differentiation

b. Only for trigonometric functions

c. Only for logarithmic integrals

d. When both functions are exponential

44 [Challenging] • Target: Cycling Integrals (Solving for the Unknown Integral) • Cross-Lesson: 3.02 Vector-Valued Functions

Evaluate $\int e^{2x}\sin x dx$.

a. $\frac{e^{2x}}{5}(2\sin x - \cos x) + C$

b. $\frac{e^{2x}}{5}(\sin x - 2\cos x) + C$

c. $\frac{e^{2x}}{4}(2\sin x - \cos x) + C$

d. $e^{2x}\cos x + C$

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45 [Challenging] • Target: Integration by Parts with Inverse Trigonometric Functions • Cross-Lesson: 4.01 Polar Curves and Derivatives

Evaluate $\int \arctan x dx$.

a. $x\arctan x - \frac{1}{2}\ln(1+x^2) + C$

b. $x\arctan x + \frac{1}{2}\ln(1+x^2) + C$

c. $\frac{1}{1+x^2} + C$

d. $x\arctan x - \ln(1+x) + C$

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